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Uptake of perfluorooctanoic acid by Caco-2 cells: Involvement of organic anion transporting polypeptides

Osamu Kimura[a,*], Yukiko Fujii[b], Koichi Haraguchi[b], Yoshihisa Kato[c], Chiho Ohta[d], Nobuyuki Koga[d], Tetsuya Endo[a]

[a] School of Pharmaceutical Sciences, Health Sciences University of Hokkaido, 1757 Kanazawa, Ishikari-Tobetsu, Hokkaido, 061-0293, Japan
[b] Daiichi College of Pharmaceutical Sciences, Minami-Ku, Fukuoka, 815-8511, Japan
[c] Kagawa School of Pharmaceutical Sciences, Tokushima Bunri University, Sanuki, Kagawa, 769-2193, Japan
[d] Faculty of Nutritional Sciences, Nakamura Gakuen University, Johnan-Ku, Fukuoka, 814-0198, Japan

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ABSTRACT

The mechanism underlying the intestinal absorption of perfluorooctanoic acid (PFOA) was investigated using Caco-2 cells. The uptake of PFOA from the apical membrane of Caco-2 cells was fast, and pH, temperature, and concentration dependent, but Na^+ independent. Coincubation with sulfobromophthalein (BSP), glibenclamide, estron-3-sulfate, cyclosporine A or rifamycin SV, which are typical substrates or inhibitors of organic anion transporting polypeptides (OATPs), significantly decreased the uptake of PFOA. However, coincubation with probenecid or *p*-aminohippuric acid, typical substrates of organic anion transporters, did not decrease the uptake of PFOA. Furthermore, coincubation with L-lactic acid or benzoic acid, substrates of monocarboxylic acid transporters, did not decrease PFOA uptake. The relationship between the initial uptake of PFOA and its concentration was saturable, suggesting the involvement of a carrier-mediated process. The calculated K_m and uptake clearance (V_{max}/K_m) values for PFOA were 8.3 μM and 55.0 μL/mg protein/min, respectively. This clearance value was about 3-fold greater than that of the non-saturable uptake clearance (K_d: 18.1 μL/mg protein/min). Lineweaver–Burk plots revealed that BSP competitively inhibits the uptake of PFOA, with a K_i value of 23.1 μM. These results suggest that the uptake of PFOA from the apical membranes of Caco-2 cells could be, at least in part, mediated by OATPs along with BSP.

1. Introduction

Perfluorooctanoic acid (PFOA) is typical of the per- and polyfluorinated chemicals used in various industrial applications. PFOA is very stable chemically and persists in the environment, and has been found in the serum of humans and wildlife species in various countries around the world (Butenhoff et al., 2004; Kannan et al., 2004; Lau et al., 2007; Calafat et al., 2007; Post et al., 2012; Gebbink et al., 2015). The biological half-life of PFOA in humans is reported to be approximately 3.8 years (Olsen et al., 2007).

Possible exposure routes of humans to PFOA include dust, daily diet, drinking water and inhaled air. The serum concentration of PFOA in the general population in the United States is low (approximately 4–5 μg/L) (Calafat et al., 2007) and the adverse health effects associated with exposure to low levels of environmental PFOA remain controversial. However, various adverse health effects including hyperuricemia (Steenland et al., 2010) and hypercholesterolemia (Steenland et al., 2009; Frisbee et al., 2010) have been reported in workers occupationally exposed to PFOA and residents consuming drinking water contaminated with PFOA from a chemical plant. Barry et al. (2013) reported that PFOA exposure was positively associated with kidney and testicular cancer in occupationally exposed workers and the general population living near a chemical plant.

In experimental animals, PFOA is rapidly and highly absorbed following oral administration, with the highest plasma concentration achieved within a few hours (Fujii et al., 2015a). However, the absorption mechanism of PFOA has not yet investigated in detail. PFOA is not metabolized, and is distributed mainly in the liver and plasma, followed by the kidney and, to a lesser extent, other tissues, and it is excreted mainly in the urine (Kennedy et al., 2004; Hundley et al., 2006; Lau et al., 2007; Lou et al., 2009; Post et al., 2012).

The human colorectal adenocarcinoma cell line Caco-2 is a useful model with which to study the intestinal absorption of various compounds as the cells are morphologically and functionally similar to human small intestinal epithelial cells (Hilgers et al., 1990). Several studies have reported that organic anion transporting polypeptides

* Corresponding author. Tel.: +81 133 23 1193; fax: +81 133 23 1193.
E-mail address: o_kimura@hoku-iryo-u.ac.jp (O. Kimura).

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(OATPs) have been detected in the apical membranes of both human small intestine and Caco-2 cells (Kobayashi et al., 2003; Sai et al., 2006; Glaeser et al., 2007) and have an important role in the intestinal absorption not only of endogenous organic anion compounds but also pharmaceutical anionic drugs, such as sulfobromophthalein (BSP), estron-3-sulfate, fexofenadine and glibenclamide (Kis et al., 2010; Visentin et al., 2012; Sai et al., 2006; Satoh et al., 2005; Roth et al., 2012; Estudante et al., 2013). Recently, Yang et al. (2009) and Weaver et al. (2010) reported that the uptake of PFOA was higher in oatp1a1-expressing CHO cells than in wild-type CHO cells. As well as OATPs, Caco-2 cells express several transporters such as monocarboxylic acid transporters (MCTs) and oligopeptide transporters (Hadjiagapiou et al., 2000; Seithel et al., 2006; Hilgendorf et al., 2007). The purpose of this study was to investigate whether PFOA uptake across the apical membranes of Caco-2 cells is mediated by OATPs and/or other transporter(s).

2. Materials and methods

2.1. Materials

Native perfluorooctanoic acid (PFOA), *p*-aminohippuric acid, cyclosporin A and Dulbecco's modified Eagle's medium (DMEM), tetrabutylammonium hydrogen sulfate and 11H-perfluoroundecanoic acid were purchased from Wako Pure Chemical Industries, Ltd. (Osaka, Japan). Glibenclamide, sulfobromophthalein (BSP), estron-3-sulfate, 4,4′-diisothiocyanostilbene-2,2′-disulfonic acid (DIDS), rifamycin SV and probenecid were purchased from Sigma–Aldrich (St. Louis, MO). Fetal bovine serum (FBS) and nonessential amino acid (NEAA) were obtained from Life Technologies Co. (Carlsbad, CA). $^{13}C_4$-labeled PFOA was purchased from Wellington Laboratories Inc. (Guelph, Canada). HPLC grade methanol and methyl tert-butyl ether (MTBE) were obtained from Kanto Chemicals Co., Ltd. (Tokyo, Japan). Benzyl bromide was obtained from the Tokyo Chemical Industry Co., Ltd. (Tokyo, Japan). All other chemicals used were of the highest purity commercially available.

2.2. Cell culture

Caco-2 cells at passage 46 were obtained from the RIKEN Cell Bank (Tsukuba, Japan), and maintained in DMEM containing FBS (10%), NEAA (1%), streptomycin (100 μg/mL) and penicillin G (100 U/mL) at 37 °C in a humidified atmosphere of 5% CO_2–95% air. The cells used were between passage 55 and 76. The culture medium was replaced three times a week, and the cells were sub-cultured once a week.

2.3. Cellular uptake of PFOA

The uptake experiment was performed as described previously (Kimura et al., 2008, 2009). Caco-2 cells were grown on 35-mm six-well culture dishes coated with rat tail collagen type I (Corning Incorporated, Tewksbury, MA) at a density of 5×10^4 cells/dish. After seeding, confluent Caco-2 cell monolayers, cultured for 2 weeks, were used in the uptake study.

The incubation medium used for the uptake study was Hanks' balanced salt solution (HBSS) containing 25 mM D-glucose and 10 mM MES (at pH 5.5, 6.0 or 6.5) or 10 mM HEPES (at pH 7.0, 7.4 or 8.0). The culture medium was removed, and the cells were preincubated at 37 °C or 4 °C for 20 min in 1.5 mL of the incubation medium (pH 7.4). After preincubation, the medium was aspirated, and the cells were incubated with 1.5 mL of fresh incubation medium containing PFOA for the designated times at the same temperature as preincubation.

To examine whether the uptake of PFOA was Na^+ dependent, NaCl in the incubation medium was replaced with equimolar KCl, and Na_2HPO_4 was omitted from the medium (Kimura et al., 2014, 2016). PFOA and a number of other compounds were dissolved in methanol or dimethyl sulfoxide, and added to the incubation medium with the final concentration of vehicles being 0.5% or lower.

After incubation with PFOA, the cell surface was quickly washed three times with ice-cold incubation medium. The cells were suspended in 1.0 mL of extraction solution (0.03 M phosphate buffer (pH 7.0):methanol = 1:1) for 60 min at room temperature, and the cells were scraped off and collected using a cell scraper (Kimura et al., 2008, 2009). The suspension was centrifuged at 13,000 × *g* for 10 min.

2.4. Determination of PFOA, BSP and protein levels

Determination of PFOA was carried out by gas chromatography–mass spectrometry (GC–MS, Agilent 6890GC/5973inertMSD, CA, USA) based on previous studies (Fujii et al., 2012, 2013, 2015b). Briefly, 1–10 μL of cell supernatant (depending on PFOA concentration) and an internal standard (0.5 ng of $^{13}C_4$-labeled PFOA) were added to a vial and then dried. The dried sample was redissolved in 100 μL of a 0.1 mol L^{-1} benzyl bromide/MTBE solution with an injection standard (11H-perfluoroundecanoic acid) and derivatized to benzyl esters at 60 °C for 1 h. The derivatized PFOA was analyzed by GC–MS with electron-capture negative ionization (GC/ECNI/MS) in selected ion monitoring mode. Injected samples were separated on an HP-5MS column (30 m × 0.25 mm i.d., 0.25-μm-thick film, Agilent Technologies, CA, USA) with helium carrier gas. Splitless injections (1 μL) were used with an injector temperature of 220 °C, and the split vent was opened after 1.5 min. The initial oven temperature was 70 °C (1.5 min), which was increased to 230 °C (20 °C min^{-1}), subsequently increased to 280 °C (4 °C min^{-1}), and then maintained at 280 °C (5 min). Methane was used as the reagent gas for chemical ionization (2 mL min^{-1}). Quantification and confirmation ions for PFOA determination were performed at 504 and 485 *m/z*, respectively. The instrumental detection limit was defined as the mass of the analyte producing a peak with a signal-to-noise ratio of three. Milli-Q water was used for the procedural blank control. The procedural blanks were analyzed with every batch of 15 samples. Standard stock solutions (2 μg/mL) of native PFOA were diluted to five working standard solutions (10, 1, 0.5, 0.1, and 0.01 ng/mL) using 100 μL of a 0.1 M benzyl bromide/acetone solution with 10 ng of 11H-perfluoroundecanoic acid. The calibration curve was linear and characterized by good correlation coefficients (> 0.99).

Determination of BSP was carried out using a HPLC system consisting of a Shimadzu LC-20Avp pump and SPD-10A UV detector (Kyoto, Japan) equipped with a Kinetex EVO C18 column (4.6 mm inside diameter × 250 mm; Phenomenex, CA). The mobile phase and the wavelength for the determination of BSP was 50 mM K_2HPO_4 buffer (pH 9.2) containing acetonitrile (70:30, v/v) and 580 nm, respectively. The column temperature and flow rate were 40 °C and 1.0 mL/min, respectively. The calibration curve of the BSP used was linear over the concentration range of 0.1–1.0 nmol/mL (r = 0.99) and the coefficient of variance was below 4%.

The protein concentration was determined using a Bio-rad dye reagent (Richmond, CA) with bovine serum albumin as the standard.

2.5. Kinetic analysis of PFOA uptake

According to previous reports (Kimura et al., 2008, 2009, 2014), the uptake rate of PFOA by Caco-2 cells (V) could be fitted to the following equation.

$$V = \frac{V_{max}[S]}{(K_m + [S])} + K_d[S] \quad (1)$$

where V_{max} is the maximum uptake rate for a carrier-mediated process, $[S]$ is the substrate concentration in the medium, K_m is the half-saturation concentration (Michaelis–Menten constant), and K_d is a coefficient of passive diffusion. The parameters V_{max}, K_m and K_d were calculated using the KaleidaGraph program (HULINKS Inc., Japan). The

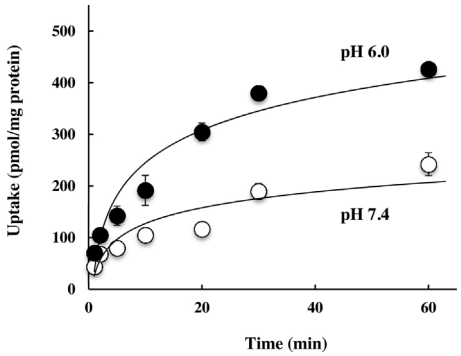

Fig. 1. Time course of the uptake of PFOA by Caco-2 cells. Caco-2 cells were incubated for the indicated periods at 37 °C with 1 μM PFOA in a medium at pH 6.0 or pH 7.4. Each point represents the mean ± S.E. of 3–6 determinations.

inhibition constant (K_i) of BSP in the PFOA uptake was also calculated using the KaleidaGraph program.

2.6. Statistical analyses

The data were analyzed by either Student's *t*-test or Sheffe's multiple comparison test after the analysis of variance using the Statcel 3 program, and differences with values of $p < 0.05$ were considered to be significant. Data are shown as the mean ± S.E.

3. Results

3.1. Uptake of PFOA by Caco-2 cells

Caco-2 cells were incubated with 1 μM PFOA at pH 6.0 or 7.4 for up to 60 min (Fig. 1). The uptake of PFOA at pH 6.0 increased rapidly, and then reached a plateau at about 30 min. The uptake of PFOA at pH 7.4 was markedly lower than that at pH 6.0.

The effect of extracellular pH on the initial uptake of PFOA in Caco-2 cells was evaluated by incubating PFOA at a pH range of pH 5.5–8.0 for 1 min (Fig. 2). The uptake of PFOA increased with decreases in extracellular pH from 7.4 to 5.5 by 1.6-fold.

The uptake of PFOA from incubation media containing different concentrations of PFOA at pH 6.0 was also investigated (Fig. 3). The relationship between the initial uptake of PFOA at 1 min and the concentration of PFOA (1–500 μM) was non-linear. These results indicated that the uptake of PFOA involves a saturable process at low concentrations and a non-saturable process at high concentrations. The kinetic parameters calculated for PFOA uptake were as follows: $K_m = 8.3 \pm 1.2$ μM, $V_{max} = 456.6 \pm 20.4$ pmol/mg protein/min, and $K_d = 18.1 \pm 0.17$ μL/mg protein/min. The uptake clearance

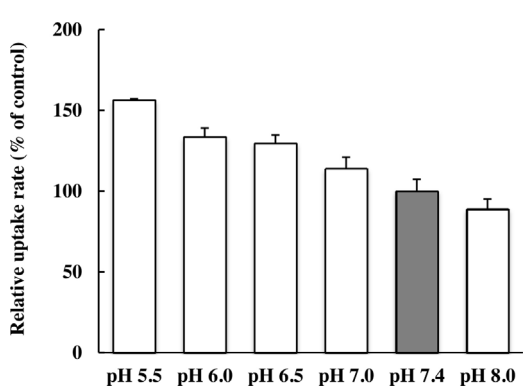

Fig. 2. Effect of extracellular pH on the uptake of PFOA by Caco-2 cells. Caco-2 cells were incubated for 1 min with 1 μM PFOA in a medium at different pH levels. Data are expressed as the percentage of the control (pH 7.4). Each point represents the mean ± S.E. of 3–6 determinations.

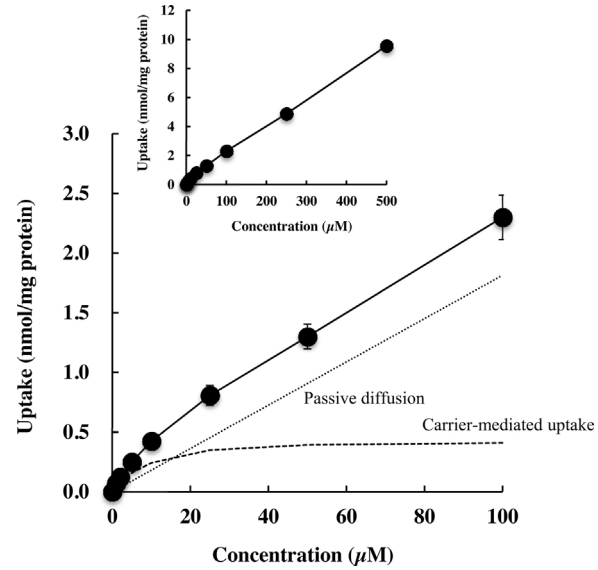

Fig. 3. Concentration dependence of the uptake of PFOA by Caco-2 cells. Caco-2 cells were incubated with various concentrations of PFOA (1–500 μM) at 37 °C and pH 6.0 for 1 min.Each point represents the mean ± S.E. for 3–6 determinations. The dotted lines indicate the uptake of PFOA via a carrier-mediated process and passive diffusion, respectively.

(V_{max}/K_m) was 55.0 μL/mg protein/min.

3.2. Effects of temperature, sodium ions and various compounds on the uptake of PFOA

The effects of low temperature, Na^+, and various compounds on the uptake of PFOA by Caco-2 cells were investigated (Table 1). Incubation at low temperature (4 °C) significantly reduced the uptake of PFOA at pH 6.0 by 54%. Na^+-free conditions, achieved by the replacement of NaCl in the incubation medium with KCl, did not affect the uptake of PFOA. Coincubation with BSP, glibenclamide, estron-3-sulfate, cyclosporine A or rifamycin SV (OATP substrates or inhibitors) significantly decreased the uptake of PFOA by about 30–40%. However, coincubation with probenecid or *p*-aminohippulic acid (typical substrates of organic anion transporters, OATs) and coincubation with L-lactic acid or benzoic acid (typical substrates of MCTs) did not reduce the uptake of PFOA. Furthermore, coincubation with DIDS (a non-specific anion exchange inhibitor) did not reduce the uptake of PFOA.

Table 1
Effects of various compounds on the uptake of PFOA by Caco-2 cells.

Compound	Relative uptake rate (% of control)
Control (37 °C and pH 6.0)	100 ± 3.9
4 °C	45.8 ± 2.4*
Na^+-free[a]	90.4 ± 3.9*
BSP (100 μM)	64.1 ± 3.5*
Glibenclamide (100 μM)	66.7 ± 2.0*
Estron-3-sulfate (100 μM)	70.5 ± 2.4*
Cyclosporine A (100 μM)	68.3 ± 3.1*
Rifamycin SV (100 μM)	60.7 ± 4.1*
DIDS (1 mM)	88.9 ± 2.3*
Probenecid (1 mM)	88.2 ± 3.2
p-Aminohippuric acid (1 mM)	102.8 ± 4.3
L-Lactic acid (5 mM)	103.8 ± 6.2
Benzoic acid (5 mM)	102.5 ± 4.3

Caco-2 cells were incubated with 1 μM PFOA at 37 °C for 1 min in the presence or absence of various compounds at pH 6.0.

[a] Caco-2 cells were incubated at 37 °C and pH 6.0 for 1 min in a Na^+-free medium.

* Significantly different from the control.

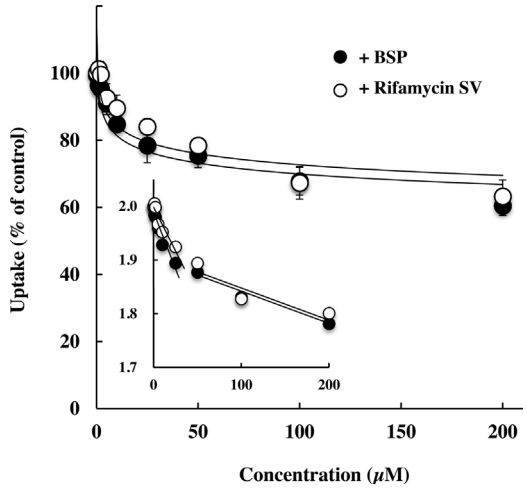

Fig. 4. Inhibitory effects of BSP and rifamycin SV on the uptake of PFOA by Caco-2 cells.Caco-2 cells were incubated with PFOA (1 μM) for 1 min at 37 °C in the presence different concentrations of BSP or rifamycin SV. The insert shows the logarithmic transformed uptake (% of control) of PFOA. Each point represents the mean ± S.E. for 3–6 determinations.

3.2.1. Concentration-dependent inhibition of PFOA uptake in Caco-2 cells by BSP and rifamycin SV

The effects of BSP and rifamycin SV on the uptake of PFOA were examined (Fig. 4). Coincubation with BSP or rifamycin SV at different concentrations (0.1–200 μM) reduced the uptake of PFOA in a concentration-dependent manner. The inhibition at 100 μM BSP and rifamycin SV was about 40%.

3.3. Competitive inhibition of BSP on the uptake of PFOA

The mode of the BSP-induced inhibition of PFOA uptake was investigated using Lineweaver–Burk plots (Fig. 5). Caco-2 cells were incubated with various concentrations of PFOA with or without 100 μM BSP. As shown in Fig. 5, BSP inhibited the uptake of PFOA in a competitive manner with an inhibition constant (K_i) value of 23.1 μM.

To confirm the uptake of PFOA via OATP, the inhibition of BSP uptake by PFOA was investigated (Fig. 6). Furthermore, the inhibition of BSP uptake by estron-3-sulfate and rifamycin SV were also investigated. Coincubation with 100 μM PFOA, estron-3-sulfate or rifamycin SV significantly decreased BSP uptake to a similar degree (about 40%). Incubation at low temperature (4 °C) significantly reduced the uptake of BSP by 53.2 ± 3.49% (n = 3) (data not shown in Fig. 6).

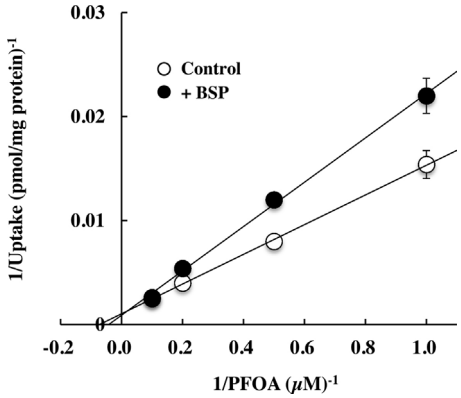

Fig. 5. Lineweaver–Burk plots for the uptake of PFOA by Caco-2 cells. Caco-2 cells were incubated with various concentrations of PFOA (1–20 μM) at 37 °C and pH 6.0 for 1 min in the presence or absence of 100 μM BSP. Each point represents the mean ± S.E. for 4–5 determinations.

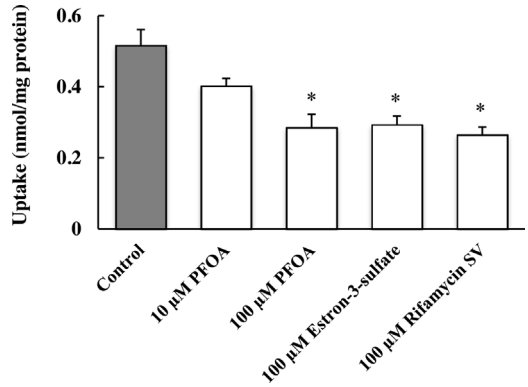

Fig. 6. Effects of PFOA, estron-3-sulfate and rifamycin SV on the uptake of BSP by Caco-2 cells. Caco-2 cells were incubated with 5 μM BSP at 37 °C and pH 6.0 for 5 min in the absence (control) or presence of PFOA, estron-3-sulfate or rifamycin SV. Each point represents the mean ± S.E. for 5–6 determinations. *Significantly different from the control.

4. Discussion

Coincubation with BSP or rifamycin SV (an inhibitor of OATPs) reduced the uptake of PFOA in a concentration-dependent manner (Fig. 4). BSP competitively inhibited PFOA uptake (Fig. 5) with an apparent K_i value of 23.1 μM, which is similar to the K_m value for the saturable uptake of PFOA (8.3 μM). As well as BSP, coincubation with other OATPs inhibitors, glibenclamide, estron-3-sulfate, cyclosporine A and rifamycin SV, significantly decreased the uptake of PFOA by 30–40% (Table 1). The uptake of PFOA from the apical membranes of Caco-2 cells was Na^+ independent (Table 1), and a decrease in extracellular pH tended to increase the initial uptake of PFOA (Fig. 2). In agreement with our results, OATPs are reported to be Na^+ independent (Roth et al., 2012; Estudante et al., 2013), and OATP activity may be enhanced at a lower extracellular pH, although this pH dependency remains somewhat controversial (Kobayashi et al., 2003; Satoh et al., 2005; Sai et al., 2006; Roth et al., 2012; Visentin et al., 2012; Estudante et al., 2013). The present results suggested that the uptake of PFOA from the apical membranes of Caco-2 cells is mediated by OATPs and shares the same transporter(s) as BSP. In agreement, Yang et al. (2009) and Weaver et al. (2010) reported that the uptake of PFOA was higher in oatp1a1-expressing CHO cells than in wild-type CHO cells. The OATP-mediated uptake of PFOA could be responsible for the rapid uptake of PFOA into Caco-2 cells found in the present study as well as the rapid and high absorption of PFOA in vivo (Kennedy et al., 2004; Hundley et al., 2006; Lau et al., 2007; Lou et al., 2009).

The initial uptake of PFOA from the apical membranes was concentration dependent (Fig. 3), and fitted the equation including carrier-mediated uptake and passive diffusion. The K_m value of the affinity site and the uptake clearance (V_{max}/K_m) values for the carrier-mediated uptake for PFOA were calculated to be 8.3 μM and 55.0 μL/mg protein/min, respectively. This clearance value (55.0 μL/mg protein/min), which indicates uptake via a carrier-mediated process, was 3.0-fold greater than that of the non-saturable uptake clearance (K_d: 18.1 μL/mg protein/min). These results suggest that PFOA is mainly taken up from the apical membranes of Caco-2 cells via a carrier-mediated process. However, the contribution of carrier-mediated process in the uptake of PFOA is estimated to be 40–50%, because of the inhibition by the low temperature and the OATP inhibitors (substrates) (Table 1). Further study is necessary to estimate the contribution of OATP in the uptake of PFOA in Caco-2 cells.

Sai et al. (2006) and Kis et al. (2010) reported that the uptake of estrone-3-sulfate (a typical substrate for OATP) in Caco-2 cells is fitted to the biphasic saturation model: The K_m values of high- and low-affinity sites were 1.81 μM and 1.4 mM, respectively (Sai et al., 2006), and 6 μM and 1.5 mM, respectively (Kis et al., 2010). These values of K_m of

the high-affinity site were similar to the K_m value of the present result of PFOA (8.3 μM). However, it remains open whether OATP at the low affinity (K_m was about 1.5 mM) contribute the uptake of PFOA in Caco-2 cells, because we investigated the uptake of PFOA in the range between 1 and 500 μM (Fig. 3). Noteworthly is that the inhibitory effects of BSP and rifamysin SV on the PFOA uptake did not fit the single straight line (Fig. 4, insert).

OATP1A2 and OATP2B1 are members of the human OATP family and are localized in the apical membranes of human small intestinal as well as Caco-2 cells (Kobayashi et al., 2003; Sai et al., 2006; Glaeser et al., 2007; Roth et al., 2012; Estudante et al., 2013). The amount of OATP2B1 expressed in human small intestinal and Caco-2 cells may be higher than that of OATP1A2 as the mRNA expression level of OATP2B1 is also higher than those of other OATP isoforms (OATP3A1, OATP4A1 and OATP 1A2) in human intestinal and Caco-2 cells (Sai et al., 2006; Meier et al., 2007; Hilgendorf et al., 2007). Yang et al. (2010) reported that OATP1A2 is not involved in the cellular uptake of PFOA as the uptake rates of PFOA did not differ significantly between OATP1A2-expessing HEK cells and mock (vector-transfected) cells. Thus, OATP2B1, rather than OAPT1A2, may mediate the uptake of PFOA in Caco-2 cells. Further studies using overexpression or siRNA of OATP2B1 techniques are necessary to clarify the OATP subtype mediating PFOA uptake.

Apart from OATPs, Caco-2 cells express various transporters such as monocarboxylic acid transporters (MCTs) (Hadjiagapiou et al., 2000) and anion exchange transporters (Ogihara et al., 1999; Kimura et al., 2011). However, coincubation with L-lactic acid or benzoic acid (typical substrates of MCTs) had no effect on the uptake of PFOA, and coincubation with DIDS (a non-specific anion exchange inhibitor) also showed no effect (Table 1). These results suggested that the uptake of PFOA from the apical membranes is not mediated via MCTs or DIDS-sensitive anion exchange transporters.

Recently, several research groups have reported that the renal secretion and reabsorption of PFOA are mediated via OATs (Nakagawa et al., 2007, 2009; Yang et al., 2010; Han et al., 2012). However, OATs are reported not to be expressed or showed very low expression levels in human intestinal and Caco-2 cells (Seithel et al., 2006; Hilgendorf et al., 2007; Estudante et al., 2013). In the present study, coincubation with *p*-aminohippuric acid and probenecid (substrates of OATs) did not affect the uptake of PFOA (Table 1). Thus, the uptake of PFOA via OATs is speculated to be negligible.

In conclusion, PFOA appears to be taken up from the apical membranes of Caco-2 cells, at least in part by OATPs along with BSP. The OATP-mediated uptake of PFOA could be responsible for the rapid and high absorption of PFOA reported previously.

Conflict of interest

The authors declare that we have no conflicts of interest to disclose.

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